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Electronic component

Abstract

An electronic component includes a component body, a metal terminal, and a bonding material. The component body includes an element body and an external electrode. The metal terminal includes a first main surface and a second main surface, and a side surface. The bonding material electrically and physically connects the external electrode and the metal terminal. The metal terminal includes a first metal layer including the first main surface, a second metal layer including the second main surface, and a terminal body including the side surface. The terminal body is exposed at the side surface, and the first metal layer and the second metal layer are separated from each other on the side surface. Each of the first metal layer and the second metal layer includes a Ni plated layer. The terminal body includes Cu. The bonding material includes solder.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2022-143135, filed on Sep. 8, 2022. The entire contents of which are incorporated herein by reference.

BACKGROUND

Field

(2) The present disclosure relates to an electronic component.

Description of the Related Art

(3) Known electronic components include a component body, a metal terminal, and a bonding material that connects the component body and the metal terminal (for example, refer to Japanese Unexamined Utility Model Publication No. H2-45620). The component body includes an element body and an external electrode disposed on the element body. The bonding material electrically and physically connects the external electrode and the metal terminal.

SUMMARY

- (4) An object of an aspect of the present disclosure is to provide an electronic component that reduces a decrease in mechanical strength of a metal terminal and reduces an increase in transmission loss due to a skin effect in a high frequency range.
- (5) An electronic component according to one aspect of the present disclosure includes a component body, a metal terminal, and a bonding material. The component body includes an element body and an external electrode disposed on the element body. The metal terminal includes a first main surface and a second main surface opposing each other, and a side surface coupling the first main surface and the second main surface. The bonding material is disposed between the external electrode and the first main surface, and electrically and physically connects the external electrode and the metal terminal. The metal terminal includes a first metal layer including the first main surface, a second metal layer including the second main surface, and a terminal body including the side surface. The terminal body is exposed at the side surface, and the first metal layer and the second metal layer are separated from each other on the side surface. Each of the first metal layer and the second metal layer includes a Ni plated layer. The terminal body includes Cu. The bonding material includes solder.
- (6) In the one aspect, each of the first and second metal layers includes a Ni plated layer. Therefore, even in the configuration in which the terminal body includes Cu and the bonding material includes the solder, solder leaching tends not to occur in the terminal body. As a result, the one aspect reduces a decrease in the mechanical strength of the metal terminal.
- (7) In the high frequency range, a skin effect occurs, in which a current tends to flow near the surface of a conductor. Ni is a magnetic material. For example, the skin depth of a conductor including Ni is smaller than the skin depth of a conductor including Cu. The conductor including Ni increases the transmission loss due to the skin effect in the high frequency range as compared with the conductor including Cu.
- (8) In the one aspect, in the high frequency range, a current tends to flow near the surface of the metal terminal. The terminal body including Cu is exposed at the side surface of the metal terminal. Ni tends not to be exposed at the side surface. Therefore, the one aspect reduces an increase in the transmission loss due to the skin effect in the high frequency range.
- (9) In the one aspect, Cu included in the terminal body may be exposed at the side surface.
- (10) The configuration in which Cu is exposed at the side surface certainly reduces an increase in the transmission loss due to the skin effect in the high frequency range.
- (11) In the one aspect, the metal terminal may include a plurality of portions opposing the external electrode, connected to the bonding material, and separated from each other. Each of the plurality of portions may include a surface exposed to a space between the plurality of portions. The side surface may include the surface included in each of the plurality of portions.
- (12) The configuration in which the metal terminal includes the plurality of portions increases the ratio of the side surface to an outer surface of the metal terminal. Therefore, the configuration more certainly reduces an increase in the transmission loss due to the skin effect in the high frequency range.
- (13) In the one aspect, the terminal body may have a thickness of 0.05 mm or more.
- (14) The configuration in which the terminal body has the above thickness still more certainly reduces an increase in the transmission loss due to the skin effect in the high frequency range.
- (15) In the one aspect, the Ni plated layer may have a thickness of 1 to 4 μm .
- (16) The configuration in which the Ni plated layer has the above thickness still more certainly reduces an increase in the transmission loss due to the skin effect in the high frequency range.
- (17) In the one aspect, the first metal layer may include a Sn plated layer including the first main surface. The second metal layer may include a Sn plated layer including the second main surface.
- (18) The configuration in which each of the first and second metal layers includes the Sn plated layer still more certainly reduces an increase in the transmission loss due to the skin effect in the high frequency range.

- (19) In the one aspect, the component body may include an internal conductor disposed in the element body and electrically and physically connected to the external electrode. The internal conductor may include a first region physically connected to the external electrode, and a second region continuous with the first region and having a width larger than the width of the first region. The first region may include an end edge that is separated from an end edge of the side surface in a width direction of the first region by a shortest distance of 0.5 to 1 mm.
- (20) The configuration in which the first region includes the end edge still more certainly reduces an increase in the transmission loss due to the skin effect in the high frequency range.
- (21) In the one aspect, the component body may include an internal conductor disposed in the element body and electrically and physically connected to the external electrode. The internal conductor may include a first region physically connected to the external electrode, and a second region continuous with the first region and having a width larger than the width of the first region. The first region may include an end edge that is separated from an end edge of the terminal body in a width direction of the first region by a shortest distance of 0.34 to 0.88 mm.
- (22) The configuration in which the first region includes the end edge still more certainly reduces an increase in the transmission loss due to the skin effect in the high frequency range.
- (23) In the one aspect, the component body may include a plurality of internal conductors disposed in the element body and opposing each other. The metal terminal may include a first portion connected to the external electrode and extending in a direction in which the plurality of internal conductors oppose each other, and a second portion extending in a direction intersecting the direction in which the plurality of internal conductors oppose each other.
- (24) The configuration in which the metal terminal includes the first and second portions allows the second portion to function as a mounting area to be connected to an electronic device.
- (25) In the one aspect, the metal terminal may include a first portion connected to the external electrode and extending in a first direction, and a second portion extending in a second direction intersecting the first direction. The side surface may include a first surface extending in the first direction in the first portion and a second surface continuous with the first surface and extending in the second direction in the second portion.
- (26) The configuration in which the side surface includes the first and second surfaces increases the ratio of the side surface to the outer surface of the metal terminal. This configuration further certainly reduces an increase in the transmission loss due to the skin effect in the high frequency range.
- (27) The present disclosure will become more fully understood from the detailed description given hereinafter and the accompanying drawings which are given by way of illustration only, and thus are not to be considered as limiting the present disclosure.
- (28) Further scope of applicability of the present disclosure will become apparent from the detailed description given hereinafter. However, it should be understood that the detailed description and specific examples, while indicating examples of the disclosure, are given by way of illustration only, since various changes and modifications within the spirit and scope of the disclosure will become apparent to those skilled in the art from this detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view illustrating an electronic component according to an example;
- (2) FIG. 2 is a diagram illustrating the cross-sectional configuration of the electronic component according to the present example;
- (3) FIG. 3 is a diagram illustrating the cross-sectional configuration of the electronic component according to the present example;

- (4) FIG. 4 is a perspective view illustrating an electronic component according to a first modification of the present example;
- (5) FIG. 5 is a perspective view illustrating an electronic component according to a second modification of the present example;
- (6) FIG. 6 is a diagram illustrating the cross-sectional configuration of the electronic component according to the second modification of the present example;
- (7) FIG. 7 is a table illustrating an equivalent series resistance (ESR) of each sample;
- (8) FIG. 8 is a table illustrating an ESR and crotch tear strength of each sample;
- (9) FIG. 9 is a table illustrating an ESR of each sample; and
- (10) FIG. 10 is a table illustrating an ESR of each sample.

DETAILED DESCRIPTION

(11) Hereinafter, examples of the present disclosure will be described in detail with reference to the accompanying drawings. In the following description, the same elements or elements having the same functions are denoted with the same reference numerals and overlapped explanation is omitted.

(12) An electronic component according to the present example will be described with reference to FIGS. 1 to 3. FIG. 1 is a perspective view illustrating an electronic component according to the present example. Each of FIG. 2 and FIG. 3 is a diagram illustrating the cross-sectional configuration of the electronic component according to the present example. In the present example, an electronic component ED1 including a multilayer ceramic capacitor will be described.

(13) The electronic component ED1 includes a component body MD1. In the present example, the component body MD1 includes a multilayer capacitor. The multilayer capacitor includes a multilayer ceramic capacitor. The component body MD1 includes an element body 1. The element body 1 includes, for example, a plurality of dielectric layers laminated in a first direction D1. In the present example, the element body 1 has a rectangular parallelepiped shape. An outer surface of the element body 1 includes a pair of end surfaces 1a and 1b opposing each other and a side surface 1c coupling the end surface 1a and the end surface 1b. The side surface 1c is adjacent to the end surfaces 1a and 1b. The side surface 1c includes a pair of side surfaces 1c1 and 1c2 opposing each other and a pair of side surfaces 1c3 and 1c4 opposing each other. The end surfaces 1a and 1b, the side surfaces 1c1 and 1c2, and the side surfaces 1c3 and 1c4 have, for example, a rectangular shape. A “rectangular parallelepiped shape” in the present specification includes a rectangular parallelepiped shape in which corner portions and ridge portions are chamfered, or a rectangular parallelepiped shape in which corner portions and the ridge portions are rounded. A “rectangular shape” in the present specification includes, for example, a shape in which each corner is chamfered or a shape in which each corner is rounded.

(14) The end surfaces 1a and 1b oppose each other in a second direction D2. The second direction D2 intersects the first direction D1. The end surfaces 1a and 1b are, for example, orthogonal to the second direction D2. The side surfaces 1c1 and 1c2 are adjacent to the end surfaces 1a and 1b and oppose each other in the first direction D1. The side surfaces 1c1 and 1c2 are, for example, orthogonal to the first direction D1. The side surfaces 1c3 and 1c4 are adjacent to the end surfaces 1a and 1b and the side surfaces 1c1 and 1c2, and oppose each other in a third direction D3. The third direction D3 intersects the first direction D1 and the second direction D2. The side surfaces 1c3 and 1c4 define both ends of the element body 1 in the third direction D3. The side surfaces 1c3 and 1c4 are, for example, orthogonal to the third direction D3. In the present example, the first direction D1, the second direction D2, and the third direction D3 are orthogonal to each other.

(15) The end surfaces 1a and 1b extend in the first direction D1 to couple the side surface 1c1 and the side surface 1c2. The end surfaces 1a and 1b extend in the third direction D3 to couple the side surface 1c3 and the side surface 1c4. The side surfaces 1c1 and 1c2 extend in the second direction D2 to couple the end surface 1a and the end surface 1b. The side surfaces 1c1 and 1c2 extend in the third direction D3 to couple the side surface 1c3 and the side surface 1c4. The side surfaces 1c3

and **1c4** extend in the second direction **D2** to couple the end surface **1a** and the end surface **1b**. The side surfaces **1c3** and **1c4** extend in the first direction **D1** to couple the side surface **1c1** and the side surface **1c2**. The end surfaces **1a** and **1b**, the side surfaces **1c1** and **1c2**, and the side surfaces **1c3** and **1c4** may be indirectly adjacent to each other. In this case, a ridge portion is located between the end surfaces **1a** and **1b**, the side surfaces **1c1** and **1c2**, and the side surfaces **1c3** and **1c4**. In FIG. 2, hatching is omitted to clearly illustrate each part.

(16) The length of the element body **1** in the first direction **D1** is, for example, 2.0 mm. The length of the element body **1** in the second direction **D2** is, for example, 5.7 mm. The length of the element body **1** in the third direction **D3** is, for example, 5.0 mm. In the element body **1**, for example, the second direction **D2** is a long side direction.

(17) In the present example, the element body **1** includes, for example, a plurality of dielectric layers. Each dielectric layer includes a part of the element body **1**. Each dielectric layer includes, for example, a ceramic sintered body. The ceramic sintered body includes, for example, a dielectric material. The dielectric material includes, for example, a BaTiO₃-based, Ba(Ti,Zr)O₃-based, (Ba,Ca)TiO₃-based, CaZrO₃-based, or (Ca,Sr)ZrO₃-based material. In the actual element body **1**, the dielectric layers are integrated to such an extent that boundaries between the dielectric layers cannot be visually recognized.

(18) The component body **MD1** includes a plurality of internal conductors **11** and **12**. In the present example, the component body **MD1** includes two internal conductors **11** and two internal conductors **12**. The internal conductors **11** and **12** oppose each other. The internal conductors **11** and **12** oppose each other in the first direction **D1**, for example. In FIG. 3, hatching is omitted to clearly illustrate each part, and, for the sake of explanation, the internal conductors **11** and **12** are intentionally shifted from each other when viewed in the first direction **D1**.

(19) The internal conductor **11** includes a region **E1a** and a region **E1b** continuous with the region **E1a**. In the present example, the region **E1b** is continuous with the region **E1a** in the first direction **D1**. The region **E1a** is exposed at the end surface **1a**. The region **E1a** is physically connected to an external electrode **21**. The region **E1b** is not exposed at the end surfaces **1a** and **1b** and the side surface **1c**. The region **E1a** and the region **E1b** have, for example, a rectangular shape when viewed in the first direction **D1**. For example, when the region **E1a** includes a first region, the region **E1b** includes a second region.

(20) The region **E1a** includes a pair of end edges **11a** and **11b** opposing each other in the third direction **D3**. The pair of end edges **11a** and **11b** defines both ends of the region **E1a** in the third direction **D3**. The end edge **11a** opposes the side surface **1c3**. The end edge **11b** opposes the side surface **1c4**. The region **E1b** includes a pair of end edges **11c** and **11d** opposing each other in the third direction **D3**. The pair of end edges **11c** and **11d** defines both ends of the region **E1b** in the third direction **D3**. The end edge **11c** opposes the side surface **1c3**. The end edge **11d** opposes the side surface **1c4**.

(21) The region **E1b** has, for example, a width larger than the width of the region **E1a**. In the present example, the width of the region **E1a** is a distance between the end edge **11a** and the end edge **11b** in the third direction **D3**. That is, the width direction of the region **E1a** is, for example, the third direction **D3**. The width of the region **E1b** is a distance between the end edge **11c** and the end edge **11d** in the third direction **D3**.

(22) The internal conductor **12** includes a region **E2a** and a region **E2b** continuous with the region **E2a**. In the present example, the region **E2b** is continuous with the region **E2a** in the first direction **D1**. The region **E2a** is exposed at the end surface **1b**. The region **E2a** is physically connected to an external electrode **22**. The region **E2b** is not exposed at the end surfaces **1a** and **1b** and the side surface **1c**. The region **E2a** and the region **E2b** have, for example, a rectangular shape when viewed in the first direction **D1**. For example, when the region **E2a** includes a first region, the region **E2b** includes a second region.

(23) The region **E2a** includes a pair of end edges **12a** and **12b** opposing each other in the third

direction D3. The pair of end edges **12a** and **12b** defines both ends of the region E2a in the third direction D3. The end edge **12a** opposes the side surface **1c3**. The end edge **12b** opposes the side surface **1c4**. The region E2b includes a pair of end edges **12c** and **12d** opposing each other in the third direction D3. The pair of end edges **12c** and **12d** defines both ends of the region E2b in the third direction D3. The end edge **12c** opposes the side surface **1c3**. The end edge **12d** opposes the side surface **1c4**.

(24) The region E2b has, for example, a width larger than the width of the region E2a. In the present example, the width of the region E2a is a distance between the end edge **12a** and the end edge **12b** in the third direction D3. That is, the width direction of the region E2a is, for example, the third direction D3. The width of the region E2b is a distance between the end edge **12c** and the end edge **12d** in the third direction D3.

(25) The internal conductors **11** and **12** include, for example, a conductive material. The conductive material included in the internal conductors **11** and **12** includes, for example, Ag, Pd, Au, Ni, Pt, or Cu. The conductive material included in the internal conductors **11** and **12** may include, for example, an Ag—Pd alloy, an Ag—Cu alloy, an Ag—Au alloy, or an Ag—Pt alloy. The internal conductors **11** and **12** are configured as, for example, a sintered body of a conductive paste including the conductive material. The internal conductor **11** and the internal conductor **12** are disposed at different positions (layers) in the first direction D1. The internal conductor **11** and the internal conductor **12** are alternately disposed, in the element body **1**, to oppose each other in the first direction D1 with an interval therebetween. The polarity of the internal conductor **11** and the polarity of the internal conductor **12** are different from each other, for example. The thickness of each of the internal conductors **11** and **12** is, for example, 0.3 to 3.0 μm .

(26) The component body MD1 includes external electrodes **21** and **22**. The external electrodes **21** and **22** are disposed on the element body **1**. The external electrodes **21** and **22** oppose each other in the second direction D2 with the element body **1** interposed therebetween.

(27) The external electrode **21** is disposed on the end surface **1a**. The external electrode **21** is disposed to cover the end surface **1a**. The external electrode **21** is also disposed on a partial surface of the side surface **1c**. The external electrode **21** covers corner portions formed of the end surface **1a** and the side surface **1c**, and ridge portions connecting the corner portions to each other. The external electrode **21** is physically connected to the region E1a of the internal conductor **11** on the end surface **1a**.

(28) The external electrode **22** is disposed on the end surface **1b**. The external electrode **22** is disposed to cover the end surface **1b**. The external electrode **22** is also disposed on a partial surface of the side surface **1c**. The external electrode **22** covers corner portions formed of the end surface **1b** and the side surface **1c**, and ridge portions connecting the corner portions to each other. The external electrode **22** is physically connected to the region E2a of the internal conductor **12** on the end surface **1b**.

(29) The external electrodes **21** and **22** include, for example, a sintered conductor layer. The sintered conductor layer is formed, for example, through sintering a conductive paste applied to the outer surface of the element body **1**. The conductive paste may include a conductive metal powder and a glass frit. The conductive paste includes, for example, a glass component, an organic binder, and an organic solvent. When the conductive paste is sintered in a reducing atmosphere, the metal powder included in the conductive paste may be oxidized. The external electrodes **21** and **22** may include a plated layer formed on the sintered conductor layer. The plated layer includes, for example, a Ni plated layer and a Sn plated layer. In the present example, the Ni plated layer is formed on the sintered conductor layer, and the Sn plated layer is formed on the Ni plated layer.

(30) The electronic component ED1 includes a metal terminal T1. The metal terminal T1 includes main surfaces **31** and **32** and side surfaces **33** and **34**. The main surface **31** is closer to the component body MD1 than the main surface **32**. The side surfaces **33** and **34** couple the main surface **31** and the main surface **32**. The side surfaces **33** and **34** oppose each other in the third

direction D3. A distance between the side surface 33 and the side surface 34 defines the length of the metal terminal T1 in the third direction D3. The side surface 33 and the side surface 34 define both ends of the metal terminal T1 in the third direction D3. For example, when the main surface 31 includes a first main surface, the main surface 32 includes a second main surface.

(31) The metal terminal T1 includes a portion P1a and a portion P1b. The portion P1a is connected to the external electrode 21 and extends in the first direction D1. The portion P1b extends in the second direction D2. The metal terminal T1 includes a portion P1c connecting the portion P1a and the portion P1b. The portion P1c extends, for example, in the first direction D1. The portion P1c is integrated with the portion P1a on the same plane, for example. The integrated portion P1a and portion P1c have a rectangular shape when viewed in the second direction D2. The portion P1a and the portion P1c are separated from each other by the length of the portion P1c when viewed in the first direction D1. The portion P1a, the portion P1c, and the portion P1b are integrally formed, for example. In the present example, an extending direction of the metal terminal T1 intersects at the boundary between the portion P1c and the portion P1b. The portion P1c and the portion P1b are, for example, orthogonal to each other. For example, when the portion P1a includes the first portion, the portion P1b includes the second portion.

(32) The metal terminal T1 includes an upper surface 35 and a lower surface 36. The upper surface 35 is located at the end edge of the portion P1a in the first direction D1. The upper surface 35 extends in the third direction D3 to couple the side surface 33 and the side surface 34. The lower surface 36 is located at the end edge of the portion P1b in the second direction D2. The lower surface 36 extends in the third direction D3 to couple the side surface 33 and the side surface 34.

(33) The electronic component ED1 includes bonding materials 2 and 4. The portion P1a opposes the external electrode 21 and is connected to the external electrode 21 through the bonding material 2. The portion P1a extends in the second direction D2 and the third direction D3. The portion P1b is connected to an electronic device M1 through the bonding material 4. The portion P1b functions as a mounting area to be connected to the electronic device M1.

(34) The bonding material 2 is disposed in the portion P1a. When viewed in the second direction D2, the outermost peripheral edge of the bonding material 2 is located inside the outermost edge of the portion P1a and is separated from the outermost edge of the portion P1a. When viewed in the second direction D2, the outermost peripheral edge of the bonding material 2 is located inside the side surfaces 33 and 34 and the upper surface 35, and is separated from the side surfaces 33 and 34 and the upper surface 35. The bonding material 4 is disposed in the portion P1b. When viewed in the first direction D1, the outermost peripheral edge of the bonding material 4 is located inside the outermost edge of the portion P1b and is separated from the outermost edge of the portion P1b. When viewed in the first direction D1, the outermost peripheral edge of the bonding material 4 is located inside the side surfaces 33 and 34 and the lower surface 36 and is separated from the side surfaces 33 and 34 and the lower surface 36.

(35) The metal terminal T1 includes a terminal body 7. The terminal body 7 includes opposing surfaces 41 and 42 opposing each other and side surfaces 33 and 34. The opposing surface 41 is closer to the component body MD1 than the opposing surface 42. The opposing surfaces 41 and 42 and the side surfaces 33 and 34 extend from the portion P1a to the portion P1b. The side surfaces 33 and 34 couple the opposing surface 41 and the opposing surface 42. The side surface 33 of the terminal body 7 coincides with the side surface 33 of the metal terminal T1. The side surface 34 of the terminal body 7 coincides with the side surface 34 of the metal terminal T1.

(36) The terminal body 7 is exposed at the side surfaces 33 and 34 of the metal terminal T1. The side surface 33 includes a surface 33a and a surface 33b. The side surface 34 includes a surface 34a and a surface 34b. The surface 33a extends in the first direction D1 in the portion P1a. The surface 33b is continuous with the surface 33a and extends in the second direction D2 in the portion P1b. The surface 34a extends in the first direction D1 in the portion P1a. The surface 34b is continuous with the surface 34a and extends in the second direction D2 in the portion P1b.

(37) The terminal body 7 includes a metal material. The metal material included in the terminal body 7 includes Cu. The terminal body 7 has a thickness of, for example, 0.05 mm or more. For example, when each of the surfaces 33a and 34a includes a first surface, each of the surfaces 33b and 34b includes a second surface.

(38) The metal terminal T1 includes a metal layer 37 and a metal layer 38. Each of the metal layers 37 and 38 includes a Ni plated layer. The Ni plated layer has a thickness of, for example, 1 to 4 μm . The metal layer 37 includes a main surface 31. The metal layer 37 is disposed on the opposing surface 41. The metal layer 37 extends from the portion P1a to the portion P1b. The metal layer 38 includes a main surface 32. The metal layer 38 is disposed on the opposing surface 42. The metal layer 38 extends from the portion P1a to the portion P1b. In the present example, none of the metal layers 37 and 38 is disposed on the side surfaces 33 and 34. Cu is exposed at the side surfaces 33 and 34. Cu exposed at the side surfaces 33 and 34 is included in the terminal body 7. The side surfaces 33 and 34 include Cu included in the terminal body 7. The metal layer 37 located on the portion P1a is connected to the bonding material 2. The metal layer 38 located on the portion P1b is connected to the bonding material 4. For example, when the metal layer 37 includes a first metal layer, the metal layer 38 includes a second metal layer.

(39) The terminal body 7 includes an upper surface 35 and a lower surface 36. In the present example, the upper surface 35 of the terminal body 7 coincides with the upper surface 35 of the metal terminal T1. The lower surface 36 of the terminal body 7 coincides with the lower surface 36 of the metal terminal T1. In the present example, Cu is exposed, for example, at the upper surface 35 and the lower surface 36. Cu exposed at the upper surface 35 and the lower surface 36 is included in the terminal body 7. A metal layer may be formed on the upper surface 35 and the lower surface 36. In this case, the metal layer formed on each of the upper surface 35 and the lower surface 36 includes, for example, a Ni plated layer. The metal layer may be formed only on one of the upper surface 35 and the lower surface 36.

(40) The metal layer 37 includes, for example, a Sn plated layer including the main surface 31. The metal layer 38 includes, for example, a Sn plated layer including the main surface 32. In the present example, the metal layer 37 includes an outermost layer 37a. The outermost layer 37a includes Sn. The metal layer 38 includes an outermost layer 38a. The outermost layer 38a includes Sn. Each of the metal layer 37 and the metal layer 38 may include a plurality of layers. In this case, the outermost layer, among the plurality of layers included in each of the metal layer 37 and the metal layer 38, includes Sn. In the metal layer 37 and the metal layer 38, another layer may be located below the layer including Sn. A layer having an electrical conductivity different from the electrical conductivity of the metal layer 37 and the metal layer 38 may be disposed between each of the metal layer 37 and the metal layer 38 and the terminal body 7. In the present example, a plating method is used as a method for forming a plated layer. The plating method includes, for example, a sputtering method, a vapor deposition method, or an electrolytic plating method.

(41) The electronic component ED1 includes a metal terminal T2. The metal terminal T2 includes main surfaces 51 and 52 and side surfaces 53 and 54. The main surface 51 is closer to the component body MD1 than the main surface 52. The side surfaces 53 and 54 couple the main surface 51 and the main surface 52. The side surfaces 53 and 54 oppose each other in the third direction D3. A distance between the side surface 53 and the side surface 54 defines the length of the metal terminal T2 in the third direction D3. The side surface 53 and the side surface 54 define both ends of the metal terminal T2 in the third direction D3. For example, when the main surface 51 includes a first main surface, the main surface 52 includes a second main surface.

(42) The metal terminal T2 includes a portion P2a and a portion P2b. The portion P2a is connected to the external electrode 22 and extends in the first direction D1. The portion P2b extends in the second direction D2. The metal terminal T2 includes a portion P2c connecting the portion P2a and the portion P2b. The portion P2c extends, for example, in the first direction D1. The portion P2c is integrated with the portion P2a on the same plane, for example. The integrated portion P2a and

portion P2c have a rectangular shape when viewed in the second direction D2. The portion P2a and the portion P2c are separated from each other by the length of the portion P2c when viewed in the first direction D1. The portion P2a, the portion P2c, and the portion P2b are integrally formed, for example. In the present example, an extending direction of the metal terminal T2 intersects at the boundary between the portion P2c and the portion P2b. The portion P2c and the portion P2b are, for example, orthogonal to each other. For example, when the portion P2a includes a first portion, the portion P2b includes a second portion.

(43) The metal terminal T2 includes an upper surface 55 and a lower surface 56. The upper surface 55 is located at the end edge of the portion P2a in the first direction D1. The upper surface 55 extends in the third direction D3 to couple the side surface 53 and the side surface 54. The lower surface 56 is located at the end edge of the portion P2b in the second direction D2. The lower surface 56 extends in the third direction D3 to couple the side surface 53 and the side surface 54.

(44) The electronic component ED1 includes bonding materials 3 and 5. The portion P2a opposes the external electrode 22 and is connected to the external electrode 22 through the bonding material 3. The portion P2a extends in the second direction D2 and the third direction D3. The portion P2b is connected to an electronic device M1 through the bonding material 5. The portion P2b functions as a mounting area to be connected to the electronic device M1. The portion P2b is located on substantially the same plane as the portion P1b.

(45) The bonding material 3 is disposed in the portion P2a. When viewed in the second direction D2, the outermost peripheral edge of the bonding material 3 is located inside the outermost edge of the portion P2a and is separated from the outermost edge of the portion P2a. When viewed in the second direction D2, the outermost peripheral edge of the bonding material 3 is located inside the side surfaces 53 and 54 and the upper surface 55, and is separated from the side surfaces 53 and 54 and the upper surface 55. The bonding material 5 is disposed in the portion P2b. When viewed in the first direction D1, the outermost peripheral edge of the bonding material 5 is located inside the outermost edge of the portion P2b and is separated from the outermost edge of the portion P2b. When viewed in the first direction D1, the outermost peripheral edge of the bonding material 5 is located inside the side surfaces 53 and 54 and the lower surface 56 and is separated from the side surfaces 53 and 54 and the lower surface 56.

(46) The metal terminal T2 includes a terminal body 8. The terminal body 8 includes opposing surfaces 61 and 62 opposing each other and side surfaces 53 and 54. The opposing surface 61 is closer to the component body MD1 than the opposing surface 62. The opposing surfaces 61 and 62 and the side surfaces 53 and 54 extend from the portion P2a to the portion P2b. The side surfaces 53 and 54 couple the opposing surface 61 and the opposing surface 62. The side surface 53 of the terminal body 8 coincides with the side surface 53 of the metal terminal T2. The side surface 54 of the terminal body 8 coincides with the side surface 54 of the metal terminal T2.

(47) The terminal body 8 is exposed at the side surfaces 53 and 54 of the metal terminal T2. The side surface 53 includes a surface 53a and a surface 53b. The side surface 54 includes a surface 54a and a surface 54b. The surface 53a extends in the first direction D1 in the portion P2a. The surface 53b is continuous with the surface 53a and extends in the second direction D2 in the portion P2b. The surface 54a extends in the first direction D1 in the portion P2a. The surface 54b is continuous with the surface 54a and extends in the second direction D2 in the portion P2b.

(48) The terminal body 8 includes a metal material. The metal material included in the terminal body 8 includes Cu. The terminal body 8 has a thickness of, for example, 0.05 mm or more. For example, when each of the surfaces 53a and 54a includes a first surface, each of the surfaces 53b and 54b includes a second surface.

(49) The metal terminal T2 includes a metal layer 57 and a metal layer 58. Each of the metal layers 57 and 58 includes a Ni plated layer. The Ni plated layer has a thickness of, for example, 1 to 4 μm . The metal layer 57 includes a main surface 51. The metal layer 57 is disposed on the opposing surface 61. The metal layer 57 extends from the portion P2a to the portion P2b. The metal layer 58

includes a main surface **52**. The metal layer **58** is disposed on the opposing surface **62**. The metal layer **58** extends from the portion **P2a** to the portion **P2b**. In the present example, none of the metal layers **57** and **58** is disposed on the side surfaces **53** and **54**. Cu is exposed at the side surfaces **53** and **54**. Cu exposed at the side surfaces **53** and **54** is included in the terminal body **8**. The side surfaces **53** and **54** include Cu included in the terminal body **8**. The metal layer **57** located on the portion **P2a** is connected to the bonding material **3**. The metal layer **58** located on the portion **P2b** is connected to the bonding material **5**. For example, when the metal layer **57** includes a first metal layer, the metal layer **58** includes a second metal layer.

(50) The terminal body **8** includes an upper surface **55** and a lower surface **56**. In the present example, the upper surface **55** of the terminal body **8** coincides with the upper surface **55** of the metal terminal **T2**. The lower surface **56** of the terminal body **8** coincides with the lower surface **56** of the metal terminal **T2**. In the present example, Cu is exposed, for example, at the upper surface **55** and the lower surface **56**. Cu exposed at the upper surface **55** and the lower surface **56** is included in the terminal body **8**. A metal layer may be formed on the upper surface **55** and the lower surface **56**. In this case, the metal layer formed on each of the upper surface **55** and the lower surface **56** includes, for example, a Ni plated layer. The metal layer may be formed only on one of the upper surface **55** and the lower surface **56**.

(51) The metal layer **57** includes, for example, a Sn plated layer including the main surface **51**. The metal layer **58** includes, for example, a Sn plated layer including the main surface **52**. In the present example, the metal layer **57** includes an outermost layer **57a**. The outermost layer **57a** includes Sn. The metal layer **58** includes an outermost layer **58a**. The outermost layer **58a** includes Sn. Each of the metal layer **57** and the metal layer **58** may include a plurality of layers. In this case, the outermost layer, among the plurality of layers included in each of the metal layer **57** and the metal layer **58**, includes Sn. In the metal layer **57** and the metal layer **58**, another layer may be located below the layer including Sn. A layer having an electrical conductivity different from the electrical conductivity of the metal layer **57** and the metal layer **58** may be disposed between each of the metal layer **57** and the metal layer **58** and the terminal body **8**.

(52) In the present example, the bonding materials **2** and **3** include solder. The bonding material **2** is disposed between the external electrode **21** and the main surface **31**, and electrically and physically connects the external electrode **21** and the metal terminal **T1**. The bonding material **3** is disposed between the external electrode **22** and the main surface **51**, and electrically and physically connects the external electrode **22** and the metal terminal **T2**. The electronic component **ED1** may include bonding materials **4** and **5** that connect the electronic component **ED1** to the electronic device **M1**. The bonding materials **4** and **5** include, for example, solder. In the present example, the electronic device **M1** includes, for example, a circuit board or other electronic component.

(53) The solder of the bonding materials **2** and **3** is formed through, for example, reflow soldering. In the present example, a solder paste is applied to predetermined positions on surfaces connected to each other with, for example, a dispenser. The solder included in the bonding materials **2** and **3** includes, for example, a Sn—Sb solder or a Pb—Sn solder. The solder of the bonding materials **4** and **5** is formed through, for example, reflow soldering. In the present example, the solder paste is applied to predetermined positions on surfaces connected to each other through, for example, a printing method. The solder included in the bonding materials **4** and **5** includes, for example, a Sn—Ag—Cu solder or a Sn—Cu solder. The solder paste includes, for example, a flux. For example, rosin is used as the flux. The solder paste may include a solvent.

(54) A positional relationship between the internal conductor **11** and the metal terminal **T1** will be described with reference to FIG. **3**. When viewed in the first direction **D1**, both ends of the metal terminal **T1** in the third direction **D3** are defined by an end edge **33c** and an end edge **34c**. The end edge **33c** is located on the outermost side of the metal terminal **T1** in the third direction **D3** on the side surface **33**. The end edge **34c** is located on the outermost side of the metal terminal **T1** in the third direction **D3** on the side surface **34**.

(55) In the present example, the end edge **11a** is located closer to the end edge **33c** than the end edge **11b**. The end edge **11a** and the end edge **33c** form a shortest distance **L1a** between each of the end edges **11a** and **11b** and the end edge **33c** in the third direction **D3**. The end edge **11b** is located closer to the end edge **34c** than the end edge **11a**. The end edge **11b** and the end edge **34c** form a shortest distance **L1b** between each of the end edges **11a** and **11b** and the end edge **34c** in the third direction **D3**. The shortest distance **L1a** and the shortest distance **L1b** may be equal to each other or may be different from each other. The shortest distances **L1a** and **L1b** are, for example, 0.5 to 1 mm.

(56) The shortest distances **L1a** and **L1b** are obtained, for example, through the following process.

(57) For example, the cross-sectional photograph of the electronic component **ED1** at a position including the regions **E1a** and **E1b** is acquired. The cross-sectional photograph includes a photograph of the cross section of the electronic component **ED1** cut along a plane orthogonal to the side surfaces **1c1** and **1c2**. The acquired cross-sectional photograph is subjected to image processing using software. On the basis of the result of the image processing, the boundary between the end edge **11a** and the end edge **33c** is determined to estimate the shortest distance **L1a**, and the boundary between the end edge **11b** and the end edge **34c** is determined to estimate the shortest distance **L1b**.

(58) Next, a positional relationship between the internal conductor **12** and the metal terminal **T2** will be described. When viewed in the first direction **D1**, both ends of the metal terminal **T1** in the third direction **D3** are defined by an end edge **53c** and an end edge **54c**. The end edge **53c** is located on the outermost side of the metal terminal **T2** in the third direction **D3** on the side surface **53**. The end edge **54c** is located on the outermost side of the metal terminal **T2** in the third direction **D3** on the side surface **54**.

(59) In the present example, the end edge **12a** is located closer to the end edge **53c** than the end edge **12b**. The end edge **12a** and the end edge **53c** give a shortest distance **L1c** between each of the end edges **12a** and **12b** and the end edge **53c** in the third direction **D3**. The end edge **12b** is located closer to the end edge **54c** than the end edge **12a**. The end edge **12b** and the end edge **54c** give a shortest distance **L1d** between each of the end edges **12a** and **12b** and the end edge **54c** in the third direction **D3**. The shortest distance **L1c** and the shortest distance **L1d** may be equal to each other or may be different from each other. The shortest distances **L1c** and **L1d** are, for example, 0.5 to 1 mm.

(60) In order to obtain the shortest distances **L1c** and **L1d**, for example, the cross-sectional photograph of the electronic component **ED1** at a position including the regions **E2a** and **E2b** is acquired. The cross-sectional photograph is a photograph of the cross section of the electronic component **ED1** cut along a plane orthogonal to the side surfaces **1c1** and **1c2**. The acquired cross-sectional photograph is subjected to image processing using software. On the basis of the result of the image processing, the boundary between the end edge **12a** and the end edge **53c** is determined to estimate the shortest distance **L1c**, and the boundary between the end edge **12b** and the end edge **54c** is determined to estimate the shortest distance **L1d**.

(61) In the present example, the shortest distance **L1a** and the shortest distance **L1c** may be equal to each other or may be different from each other. The shortest distance **L1a** and the shortest distance **L1d** may be equal to each other or may be different from each other. The shortest distance **L1b** and the shortest distance **L1c** may be equal to each other or may be different from each other. The shortest distance **L1b** and the shortest distance **L1d** may be equal to each other or may be different from each other. The shortest distances **L1a**, **L1b**, **L1c**, and **L1d** may be equal to each other or may be different from each other.

(62) The configuration of an electronic component **ED1** according to a first modification of the present example will be described with reference to FIG. 4. FIG. 4 is a perspective view of an electronic component **ED1** according to a first modification of the present example. In the present modification, two electronic component main bodies **MD1** are disposed in the first direction **D1**. In

the metal terminals **T1** and **T2** according to the first modification, in order to connect the two electronic component main bodies **MD1**, the length of the portion **P1a** in the first direction **D1** is larger than the length of the portion **P1a** according to the present example in the first direction **D1**. Each of the two electronic component main bodies **MD1** is connected to the metal terminal **T1** through the bonding material **2**, and is connected to the metal terminal **T2** through the bonding material **3**.

(63) The metal terminal **T1** according to the first modification includes the same configuration as the configuration of the metal terminal **T1** according to the present example except that the length of the portion **P1a** in the first direction **D1** is different. The metal terminal **T2** according to the first modification includes the same configuration as the configuration of the metal terminal **T2** according to the present example except that the length of the portion **P2a** in the first direction **D1** is different. The metal terminal **T1** and the metal terminal **T2** according to the first modification include, for example, the same configuration. The material of the metal terminals **T1** and **T2** according to the first modification is, for example, the same as the material of the metal terminals **T1** and **T2** according to the present example.

(64) The metal terminal **T1** includes main surfaces **31** and **32** and side surfaces **33** and **34**. The side surfaces **33** and **34** extend from the portion **P1a** to the portion **P1b**. For example, Cu is exposed at the side surfaces **33** and **34**, and the upper surface **35** and the lower surface **36**. A metal layer may be formed on the upper surface **35** and the lower surface **36**. The metal terminal **T2** includes main surfaces **51** and **52** and side surfaces **53** and **54**. The side surfaces **53** and **54** extend from the portion **P2a** to the portion **P2b**. For example, Cu is exposed at the side surfaces **53** and **54**, and the upper surface **55** and the lower surface **56**. A metal layer may be formed on the upper surface **55** and the lower surface **56**.

(65) The configuration of an electronic component **ED1** according to a second modification of the present example will be described with reference to FIGS. 5 and 6. FIG. 5 is a perspective view of an electronic component according to a second modification of the present example. FIG. 6 is a diagram illustrating the cross-sectional configuration of the electronic component according to the second modification of the present example.

(66) In the second modification, the shape of the metal terminal **T1** is different from the shape of the metal terminal **T1** of the present example. A recess **39** is formed in the metal terminal **T1**. The metal terminal **T1** according to the second modification includes the same configuration as the configuration of the metal terminal **T1** according to the present example except that the recess **39** is formed. The metal terminal **T1** and the metal terminal **T2** according to the second modification include, for example, the same configuration. The material of the metal terminals **T1** and **T2** according to the second modification is, for example, the same as the material of the metal terminals **T1** and **T2** according to the present example.

(67) The metal terminal **T1** includes a plurality of portions **P3a** and **P3b**. The portions **P3a** and **P3b** are formed in the portion **P1a**. The portions **P3a** and **P3b** are separated from each other in the third direction **D3** due to the recess **39**. The portions **P3a** and **P3b** oppose each other in the third direction **D3** with the recess **39** interposed therebetween. The portion **P3a**, the recess **39**, and the portion **P3b** are disposed in the third direction **D3** in the order of the portion **P3a**, the recess **39**, and the portion **P3b**. In FIG. 6, hatching is omitted to clearly illustrate each part, and, for the sake of explanation, a plurality of internal conductors are intentionally shifted from each other when viewed in the first direction **D1**.

(68) The metal terminal **T1** includes main surfaces **31** and **32** and side surfaces **33** and **34**. The side surfaces **33** and **34** couple the main surface **31** and the main surface **32**. The side surfaces **33** and **34** oppose each other in the third direction **D3**. The metal terminal **T1** includes a pair of upper surfaces **35p** and **35q** and a lower surface **36**. In the second modification, the metal terminal **T1** includes side surfaces **33p** and **34p** and a surface **35r** that define the recess **39**. The side surface **33p** is included in the portion **P3a**. The side surface **34p** is included in the portion **P3b**. The surface **35r** is

exposed to a space between the portion P3a and the portion P3b.

(69) The pair of upper surfaces 35p and 35q is located at the end edge of the portion P1a in the first direction D1. The upper surface 35p extends in the third direction D3 to couple the side surface 33 and the side surface 33p. The upper surface 35q extends in the third direction D3 to couple the side surface 34 and the side surface 34p. The side surface 33p and the side surface 34p oppose each other in the third direction D3 with the recess 39 interposed therebetween. The side surface 33p extends in the first direction D1 to couple the upper surface 35p and the surface 35r. The side surface 34p extends in the first direction D1 to couple the upper surface 35q and the surface 35r to each other. The surface 35r extends in the third direction D3 to couple the side surface 33p and the side surface 34p. The surface 35r defines the deepest portion of the recess 39. The lower surface 36 is located at the end edge of the portion P1b in the second direction D2. The lower surface 36 extends in the third direction D3 to couple the side surface 33 and the side surface 34. The side surface 33, the upper surface 35p, and the side surface 33p define the outer periphery of the portion P3a. The side surface 34, the upper surface 35q, and the side surface 34p define the outer periphery of the portion P3b.

(70) The electronic component ED1 according to the second modification includes two bonding materials 2. Each of the bonding materials 2 includes solder. One bonding material 2 is disposed in the portion P3a. The portion P3a is connected to the external electrode 21 through the one bonding material 2. The portion P3a opposes the external electrode 21 through the one bonding material 2 interposed therebetween. When viewed in the second direction D2, the outermost peripheral edge of the one bonding material 2 is located inside the outermost edge of the portion P3a and is separated from the outermost edge of the portion P3a. When viewed in the second direction D2, the outermost peripheral edge of the one bonding material 2 is located inside the side surface 33, the upper surface 35p, and the side surface 33p, and is separated from the side surface 33, the upper surface 35p, and the side surface 33p.

(71) Other bonding material 2 is disposed in the portion P3b. The portion P3b is connected to the external electrode 21 through the other bonding material 2. The portion P3b opposes the external electrode 21 with the other bonding material 2 interposed therebetween. When viewed in the second direction D2, the outermost peripheral edge of the other bonding material 2 is located inside the outermost edge of the portion P3b and is separated from the outermost edge of the portion P3b. When viewed in the second direction D2, the outermost peripheral edge of the other bonding material 2 is located inside the side surface 34, the upper surface 35q, and the side surface 34p, and is separated from the side surface 34, the upper surface 35p, and the side surface 34p.

(72) The electronic component ED1 according to the second modification includes a bonding material 4. The portion P1b is connected to an electronic device M1 through a bonding material 4.

(73) The metal terminal T2 includes a plurality of portions P4a and P4b. The portions P4a and P4b are formed in the portion P2a. The portions P4a and P4b are separated from each other in the third direction D3 due to the recess 59. The portions P4a and P4b oppose each other in the third direction D3 with the recess 59 interposed therebetween. The portion P4a, the recess 59, and the portion P4b are disposed in the third direction D3 in the order of the portion P4a, the recess 59, and the portion P4b.

(74) The metal terminal T2 includes main surfaces 51 and 52 and side surfaces 53 and 54. The side surfaces 53 and 54 couple the main surface 51 and the main surface 52. The side surfaces 53 and 54 oppose each other in the third direction D3. The metal terminal T1 includes a pair of upper surfaces 55p and 55q and a lower surface 56. In the second modification, the metal terminal T2 includes side surfaces 53p and 54p and a surface 55r that define the recess 59. The side surface 53p is included in the portion P4a. The side surface 54p is included in the portion P4b. The surface 55r is exposed to a space between the portion P4a and the portion P4b.

(75) The pair of upper surfaces 55p and 55q is located at the end edge of the portion P2a in the first direction D1. The upper surface 55p extends in the third direction D3 to couple the side surface 53

and the side surface **53p**. The upper surface **55q** extends in the third direction **D3** to couple the side surface **54** and the side surface **54p**. The side surface **53p** and the side surface **54p** oppose each other in the third direction **D3** with the recess **59** interposed therebetween. The side surface **53p** extends in the first direction **D1** to couple the upper surface **55p** and the surface **55r**. The side surface **54p** extends in the first direction **D1** to couple the upper surface **55q** and the surface **55r** to each other. The surface **55r** extends in the third direction **D3** to couple the side surface **53p** and the side surface **54p**. The surface **55r** defines the deepest portion of the recess **59**. The lower surface **56** is located at the end edge of the portion **P2b** in the second direction **D2**. The lower surface **56** extends in the third direction **D3** to couple the side surface **53** and the side surface **54**. The side surface **53**, the upper surface **55p**, and the side surface **53p** define the outer periphery of the portion **P4a**. The side surface **54**, the upper surface **55q**, and the side surface **54p** define the outer periphery of the portion **P4a**.

(76) The electronic component **ED1** according to the second modification includes two bonding materials **3**. Each of the bonding materials **3** includes solder. One bonding material **3** is disposed in the portion **P4a**. The portion **P4a** is connected to the external electrode **22** through the one bonding material **3**. The portion **P4a** opposes the external electrode **22** through the one bonding material **3** interposed therebetween. When viewed in the second direction **D2**, the outermost peripheral edge of the one bonding material **3** is located inside the outermost edge of the portion **P4a** and is separated from the outermost edge of the portion **P4a**. When viewed in the second direction **D2**, the outermost peripheral edge of the one bonding material **3** is located inside the side surface **53**, the upper surface **55p**, and the side surface **53p**, and is separated from the side surface **53**, the upper surface **55p**, and the side surface **53p**.

(77) The other bonding material **3** is disposed in the portion **P4b**. The portion **P4b** is connected to the external electrode **22** through the other bonding material **3**. The portion **P4b** opposes the external electrode **22** through the other bonding material **3** interposed therebetween. When viewed in the second direction **D2**, the outermost peripheral edge of the other bonding material **3** is located inside the outermost edge of the portion **P4b** and is separated from the outermost edge of the portion **P4b**. When viewed in the second direction **D2**, the outermost peripheral edge of the other bonding material **3** is located inside the side surface **54**, the upper surface **55q**, and the side surface **54p**, and is separated from the side surface **54**, the upper surface **55q**, and the side surface **54p**.

(78) The electronic component **ED1** according to the second modification includes the bonding material **5**. The portion **P2b** is connected to an electronic device **M1** through a bonding material **5**.

(79) A positional relationship between the internal conductor **11** and the metal terminal **T1** will be described with reference to FIG. **6**. When viewed in the first direction **D1**, both ends of the metal terminal **T1** in the third direction **D3** are defined by an end edge **33c** and an end edge **34c**. The end edge **33c** is located on the outermost side of the metal terminal **T1** in the third direction **D3** on the side surface **33**. The end edge **34c** is located on the outermost side of the metal terminal **T1** in the third direction **D3** on the side surface **34**. In the second modification, the end edge **33q** is located on the outermost side of the metal terminal **T1** in the third direction **D3** on the side surfaces **33p**. The end edge **34q** is located on the outermost side of the metal terminal **T1** in the third direction **D3** on the side surface **34p**.

(80) In the second modification, the end edge **11a** is located closer to the end edge **33c** than the end edge **11b**. The end edge **11a** and the end edge **33c** give a shortest distance **L1a** between each of the end edges **11a** and **11b** and the end edge **33c** in the third direction **D3**. The end edge **11b** is located closer to the end edge **34c** than the end edge **11a**. The end edge **11b** and the end edge **34c** form a shortest distance **L1b** between each of the end edges **11a** and **11b** and the end edge **34c** in the third direction **D3**. The end edge **11a** is located closer to the end edge **33q** than the end edge **11b**. The end edge **11a** and the end edge **33q** give a shortest distance **L3a** between each of the end edges **11a** and **11b** and the end edge **33q** in the third direction **D3**. The end edge **11b** is located closer to the end edge **34q** than the end edge **11a**. The end edge **11b** and the end edge **34q** give a shortest

distance **L3b** between each of the end edges **11a** and **11b** and the end edge **34q** in the third direction **D3**.

(81) In order to obtain the shortest distances **L1a**, **L1b**, **L3a**, and **L3b**, for example, the cross-sectional photograph of the electronic component **ED1** at a position including the regions **E1a** and **E1b** is acquired. The cross-sectional photograph is a photograph of the cross section of the electronic component **ED1** cut along a plane orthogonal to the side surfaces **1c1** and **1c2**. The acquired cross-sectional photograph is subjected to image processing using software. On the basis of the result of the image processing, the boundary between the end edge **11a** and the end edge **33c** is determined to calculate the shortest distance **L1a**. The boundary between the end edge **11b** and the end edge **34c** is determined to calculate the shortest distance **L1b**. The boundary between the end edge **11a** and the end edge **33q** is determined to calculate the shortest distance **L3a**. The boundary between the end edge **11b** and the end edge **34q** is determined to calculate the shortest distance **L3b**.

(82) The shortest distance **L1a** and the shortest distance **L1b** may be equal to each other or may be different from each other. The shortest distance **L3a** and the shortest distance **L3b** may be equal to each other or may be different from each other. In the second modification, the shortest distance **L1a** is larger than the shortest distance **L3a**, and the shortest distance **L1b** is larger than the shortest distance **L3b**. The shortest distances **L1a** and **L1b** are, for example, 0.5 to 1 mm. The shortest distances **L3a** and **L3b** are, for example, 0.34 to 0.88 mm.

(83) Next, a positional relationship between the internal conductor **12** and the metal terminal **T2** will be described. When viewed in the first direction **D1**, both ends of the metal terminal **T2** in the third direction **D3** are defined by an end edge **53c** and an end edge **54c**. The end edge **53c** is located on the outermost side of the metal terminal **T2** in the third direction **D3** on the side surface **53**. The end edge **54c** is located on the outermost side of the metal terminal **T2** in the third direction **D3** on the side surface **54**. In the second modification, the end edge **53q** is located on the outermost side of the metal terminal **T2** in the third direction **D3** on the side surface **53p**. The end edge **54q** is located on the outermost side of the metal terminal **T2** in the third direction **D3** on the side surface **54p**.

(84) In the second modification, the end edge **12a** is located closer to the end edge **53c** than the end edge **12b**. The end edge **12a** and the end edge **53c** give a shortest distance **L1c** between each of the end edges **12a** and **12b** and the end edge **53c** in the third direction **D3**. The end edge **12b** is located closer to the end edge **54c** than the end edge **12a**. The end edge **12b** and the end edge **54c** give a shortest distance **L1d** between each of the end edges **12a** and **12b** and the end edge **54c** in the third direction **D3**. The end edge **12a** is located closer to the end edge **53q** than the end edge **12b**. The end edge **12a** and the end edge **53q** give a shortest distance **L3c** between each of the end edges **12a** and **12b** and the end edge **53q** in the third direction **D3**. The end edge **12b** is located closer to the end edge **54q** than the end edge **12a**, and the end edge **12b** and the end edge **54q** give a shortest distance **L3d** between each of the end edges **12a** and **12b** and the end edge **54q** in the third direction **D3**.

(85) In order to obtain the shortest distances **L1c**, **L1d**, **L3c**, and **L3d**, for example, the cross-sectional photograph of the electronic component **ED1** at a position including the regions **E2a** and **E2b** is acquired. The cross-sectional photograph is a photograph of the cross section of the electronic component **ED1** cut along a plane orthogonal to the side surfaces **1c1** and **1c2**. The acquired cross-sectional photograph is subjected to image processing using software. On the basis of the result of the image processing, the boundary between the end edge **12a** and the end edge **53c** is determined to calculate the shortest distance **L1c**. The boundary between the end edge **12b** and the end edge **54c** is determined to calculate the shortest distance **L1d**. The boundary between the end edge **12a** and the end edge **53q** is determined to calculate the shortest distance **L3c**. The boundary between the end edge **12b** and the end edge **54q** is determined to calculate the shortest distance **L3d**.

(86) The shortest distance **L1c** and the shortest distance **L1d** may be equal to each other or may be

different from each other. The shortest distance **L3c** and the shortest distance **L3d** may be equal to each other or may be different from each other. In the second modification, the shortest distance **L1c** is larger than the shortest distance **L3c**, and the shortest distance **L1d** is larger than the shortest distance **L3d**. The shortest distances **L1c** and **L1d** are, for example, 0.5 to 1 mm. The shortest distances **L3c** and **L3d** are, for example, 0.34 to 0.88 mm.

(87) The shortest distances **L1a**, **L1b**, **L1c**, and **L1d** may be equal to each other or may be different from each other. The shortest distance **L3a** and the shortest distance **L3c** may be equal to each other or may be different from each other. The shortest distance **L3a** and the shortest distance **L3d** may be equal to each other or may be different from each other. The shortest distance **L3b** and the shortest distance **L3c** may be equal to each other or may be different from each other. The shortest distance **L3b** and the shortest distance **L3d** may be equal to each other or may be different from each other. The shortest distances **L3a**, **L3b**, **L3c**, and **L3d** may be equal to each other or different from each other.

(88) Next, the thicknesses of the terminal main bodies **7** and **8** will be described in detail.

(89) In order to study the ranges of the thicknesses of the terminal main bodies **7** and **8**, the present inventors conducted the following test. That is, the present inventors prepared samples **S1** to **S4** including the terminal main bodies **7** and **8** having thicknesses different from each other, and confirmed an equivalent series resistance (ESR) of each of the samples **S1** to **S4**. The results are illustrated in FIG. 7. FIG. 7 is a table illustrating the ESR of each sample.

(90) The ESR is determined as follows.

(91) For example, ESR (Ω) at the self-resonance frequency of each of the samples **S1** to **S4** is measured. An impedance analyzer is used to measure the ESR. As the impedance analyzer, for example, 4294A manufactured by Agilent Technologies is used. The samples **S1** to **S4** are mounted on a substrate for ESR measurement. The impedance analyzer is electrically connected to each of the samples **S1** to **S4**, and the ESR (Ω) of each of the samples **S1** to **S4** is measured. A high-frequency signal having a frequency range of 10 kHz or more is applied to each of the samples **S1** to **S4**. In consideration of actual use, the ESR needs to be 0.01 Ω or less. The ESR of 0.01 Ω or less tends not to increase the transmission loss due to the skin effect of the metal terminals **T1** and **T2** in the high frequency range.

(92) In the sample **S1**, the ESR was more than 0.01 Ω . In the samples **S2** to **S4**, the ESR was 0.01 Ω or less.

(93) Next, the thickness of the Ni plated layer included in the metal layers **37** and **57** and the metal layers **38** and **58** will be described in detail.

(94) In order to study the range of the thickness of the Ni plated layer, the present inventors conducted the following test. That is, the present inventors prepared samples **S5** to **S9** including the Ni plated layers having thicknesses different from each other, and confirmed the ESR and crotch tear strength of each of the samples **S5** to **S9**.

(95) A Ni plated layer was formed through a plating method, and the thickness of the Ni plated layer was controlled due to changing a time for forming plating. In each of the samples **S5** to **S9**, two samples each including a Ni plated layer having the same thickness were prepared, the ESR was confirmed for one sample, and the crotch tear strength was confirmed for the remaining samples.

(96) The results are illustrated in FIG. 8. FIG. 8 is a table illustrating an ESR and crotch tear strength of each sample.

(97) The ESR is determined for the samples **S5** to **S9** through the same test as the test for the thicknesses of the terminal main bodies **7** and **8**. In consideration of actual use, the ESR needs to be 0.01 Ω or less as described above.

(98) The crotch tear strength is determined as follows.

(99) For example, the crotch tear strength (N) of each of the samples **S5** to **S9** is measured. A tensile tester is used to measure the crotch tear strength. In a state where the metal terminal **T1** is

fixed, the metal terminal T2 is pulled in a direction away from the metal terminal T1. For each of the samples S5 to S9, tensile strength (N) when the metal terminal T2 is damaged is measured. The tensile strength (N) is defined as the crotch tear strength (N). The breakage of the metal terminal T2 includes, for example, peeling of the metal terminal T2 from the component body MD1.

(100) In consideration of actual use, the crotch tear strength needs to be 20 N or more. When the crotch tear strength is 20 N or more, the metal terminals T1 and T2 exhibit high mechanical strength.

(101) In the sample S5, the ESR was 0.01Ω or less, and the crotch tear strength was less than 20 N. In the samples S6 to S8, the ESR was 0.01Ω or less, and the crotch tear strength was 20 N or more. In the sample S9, the ESR was more than 0.01Ω, and the crotch tear strength was 20 N or more.

(102) Next, the shortest distances L1a, L1b, L1c, and L1d will be described in detail.

(103) In order to study the ranges of the shortest distances L1a, L1b, L1c, and L1d, the present inventors conducted the following test. That is, the present inventors prepared samples S10 to S12 having shortest distances L1a different from each other, and confirmed the ESR of each of the samples S10 to S12.

(104) The results are illustrated in FIG. 9. FIG. 9 is a table illustrating an ESR of each sample; and Samples having different shortest distances L1a, L1b, L1c, and L1d are prepared, for example, due to changing the lengths of the portions P1a and P2a in the third direction D3.

(105) The ESR is determined due to selecting a specimen for each of the samples S10 to S12, and subjecting each specimen to the same test as the test for the thicknesses of the terminal main bodies 7 and 8. In consideration of actual use, the ESR needs to be 0.01Ω or less.

(106) In the samples S10 to S11, the ESR was 0.01Ω or less. In the sample S12, the ESR was more than 0.01Ω.

(107) Next, the shortest distances L3a, L3b, L3c, and L3d will be described in detail.

(108) In order to study the ranges of the shortest distances L3a, L3b, L3c, and L3d, the present inventors conducted the following test. That is, the present inventors prepared samples S13 to S15 having shortest distances L3a different from each other, and confirmed the ESR of each of the samples S13 to S15.

(109) The results are illustrated in FIG. 10. FIG. 10 is a table illustrating an ESR of each sample. The samples having different shortest distances L3a, L3b, L3c, and L3d are prepared, for example, due to changing the length of at least any one of the portions P3a, P3b, P4a, and P4b and the regions E1a and E2a in the third direction D3.

(110) The ESR is determined due to selecting a specimen for each of the samples S13 to S15, and subjecting each specimen to the same test as the test for the thicknesses of the terminal main bodies 7 and 8. In consideration of actual use, the ESR needs to be 0.01Ω or less.

(111) In the samples S13 to S14, the ESR was 0.01Ω or less. In the sample S15, the ESR was more than 0.01Ω.

(112) As described above, the electronic component ED1 includes the component body MD1, the metal terminals T1 and T2, and the bonding materials 2 and 3.

(113) In the electronic component ED1, each of the metal layers 37, 38, 57, and 58 includes a Ni plated layer. Therefore, even in a configuration in which the terminal main bodies 7 and 8 include Cu and the bonding materials 2 and 3 include solder, solder leaching tends not to occur in the terminal main bodies 7 and 8. As a result, the electronic component ED1 reduces a decrease in the mechanical strength of the metal terminals T1 and T2.

(114) In the high frequency range, a skin effect occurs, in which a current tends to flow near the surface of the metal terminal Ni is a magnetic material. For example, the skin depth of a conductor including Ni is smaller than the skin depth of a conductor including Cu. The conductor including Ni increases the transmission loss due to the skin effect in the high frequency range as compared with the conductor including Cu.

(115) In the electronic component ED1, in the high frequency range, a current tends to flow near

the surfaces of the metal terminals **T1** and **T2**. The terminal body **7** including Cu is exposed at the side surfaces **33** and **34**, and the terminal body **8** including Cu is exposed at the side surfaces **53** and **54**. Ni tends not to be exposed at the side surfaces **33**, **34**, **53**, and **54**. Therefore, the electronic component **ED1** reduces an increase in the transmission loss due to the skin effect in the high frequency range.

(116) In the electronic component **ED1**, Cu included in the terminal body **7** is exposed at the side surfaces **33** and **34**, and Cu included in the terminal body **8** is exposed at the side surfaces **53** and **54**.

(117) The electronic component **ED1** certainly reduces an increase in the transmission loss due to the skin effect in the high frequency range.

(118) In the electronic component **ED1**, the metal terminal **T1** includes a plurality of portions **P3a** and **P3b** opposing the external electrode **21**, connected to the bonding material **2**, and separated from each other. The metal terminal **T2** includes a plurality of portions **P4a** and **P4b** opposing the external electrode **22**, connected to the bonding material **3**, and separated from each other.

(119) In the electronic component **ED1**, the ratio of the side surfaces **33**, **34**, **33p**, and **34p** and the surface **35r** to the outer surface of the metal terminal **T1** increases. The ratio of the side surfaces **53**, **54**, **53p**, and **54p** and the surface **55r** to the outer surface of the metal terminal **T2** increases.

Therefore, the electronic component **ED1** more certainly reduces an increase in the transmission loss due to the skin effect in the high frequency range.

(120) In the electronic component **ED1**, the terminal main bodies **7** and **8** have a thickness of 0.05 mm or more.

(121) The electronic component **ED1** still more certainly reduces an increase in the transmission loss due to the skin effect in the high frequency range.

(122) In the electronic component **ED1**, the Ni plated layer has a thickness of 1 to 4 μm .

(123) The electronic component **ED1** still more certainly reduces an increase in the transmission loss due to the skin effect in the high frequency range.

(124) In the electronic component **ED1**, the metal layers **37**, **38**, **57**, and **58** include a Sn plated layer.

(125) The electronic component **ED1** still more certainly reduces an increase in the transmission loss due to the skin effect in the high frequency range.

(126) In the electronic component **ED1**, the shortest distances **L1a**, **L1b**, **L1c**, and **L1d** are 0.5 to 1 mm.

(127) The electronic component **ED1** still more certainly reduces an increase in the transmission loss due to the skin effect in the high frequency range.

(128) In the electronic component **ED1**, the shortest distances **L3a**, **L3b**, **L3c**, and **L3d** are 0.34 to 0.88 mm.

(129) The electronic component **ED1** still more certainly reduces an increase in the transmission loss due to the skin effect in the high frequency range.

(130) In the electronic component **ED1**, the metal terminal **T1** includes the portion **P1b**, and the metal terminal **T2** includes the portion **P2b**.

(131) In the electronic component **ED1**, the portions **P1b** and **P2b** function as a mounting area to be connected to the electronic device **M1**.

(132) In the electronic component **ED1**, the side surfaces **33** and **34** include surfaces **33a** and **34a** and surfaces **33b** and **34b**. The side surfaces **53** and **54** include surfaces **53a** and **54a** and surfaces **53b** and **54b**.

(133) The electronic component **ED1** increases the ratio of the side surfaces **33** and **34** to the outer surface of the metal terminal **T1**. The electronic component **ED1** increases the ratio of the side surfaces **53** and **54** to the outer surface of the metal terminal **T2**. The electronic component **ED1** still more certainly reduces an increase in the transmission loss due to the skin effect in the high frequency range.

(134) Although the examples and modifications of the present disclosure have been described above, the present disclosure is not necessarily limited to the above-described examples and modifications can be made without departing from the gist thereof.

(135) In the electronic component ED1 illustrated in FIGS. 5 and 6, the shortest distance L1a may be equal to or less than the shortest distance L3a. The shortest distance L1b may be equal to or less than the shortest distance L3b.

(136) In the present examples and modifications, the electronic component ED1 has been described as the multilayer ceramic capacitor, but the electronic component to which the present disclosure can be applied is not limited to the multilayer ceramic capacitor. The electronic component to which the present disclosure can be applied include, for example, a multilayer varistor, a multilayer piezoelectric actuator, a multilayer thermistor, and a multilayer solid state battery.

Claims

1. An electronic component comprising: a component body including an element body and an external electrode disposed on the element body; a metal terminal including a first main surface and a second main surface opposing each other, and a side surface coupling the first main surface and the second main surface; and a bonding material disposed between the external electrode and the first main surface and electrically and physically connecting the external electrode and the metal terminal, wherein: the metal terminal includes a first metal layer including the first main surface, a second metal layer including the second main surface, and a terminal body including the side surface, and includes: a first portion connected to the external electrode and extending in a first direction, a second portion extending in a second direction intersecting the first direction, and a third portion continuous with the first portion and the second portion and connecting the first portion and the second portion; the side surface includes: a first surface included in the first portion and extending in the first direction, a second surface included in the second portion and extending in the second direction, and a third surface included in the third portion and continuous with the first surface and the second surface; the terminal body is exposed at an entirety of the first surface, the second surface, and the third surface, and the first metal layer and the second metal layer are separated from each other on the entirety of the first surface, the second surface, and the third surface; each of the first metal layer and the second metal layer includes a Ni plated layer; the terminal body includes Cu; and the bonding material includes solder.
2. The electronic component according to claim 1, wherein Cu included in the terminal body is exposed at the side surface.
3. The electronic component according to claim 1, wherein: the metal terminal includes a plurality of portions opposing the external electrode, connected to the bonding material, and separated from each other; each of the plurality of portions includes a surface exposed to a space between the plurality of portions; and the side surface includes the surface included in each of the plurality of portions.
4. The electronic component according to claim 1, wherein the terminal body has a thickness of 0.05 mm or more.
5. The electronic component according to claim 1, wherein the Ni plated layer has a thickness of 1 to 4 μm .
6. The electronic component according to claim 1, wherein: the first metal layer includes a Sn plated layer including the first main surface; and the second metal layer includes a Sn plated layer including the second main surface.
7. The electronic component according to claim 1, wherein: the component body further includes an internal conductor disposed in the element body and electrically and physically connected to the external electrode; the internal conductor includes a first region physically connected to the external electrode, and a second region continuous with the first region and having a width larger

than the width of the first region; and the first region includes an end edge that is separated from an end edge of the side surface in a width direction of the first region by a shortest distance of 0.5 to 1 mm.

8. The electronic component according to claim 1, wherein: the component body further includes an internal conductor disposed in the element body and electrically and physically connected to the external electrode; the internal conductor includes a first region physically connected to the external electrode, and a second region continuous with the first region and having a width larger than the width of the first region; and the first region includes an end edge that is separated from an end edge of the terminal body in a width direction of the first region by a shortest distance of 0.34 to 0.88 mm.

9. The electronic component according to claim 1, wherein: the component body further includes a plurality of internal conductors disposed in the element body and opposing each other; and the metal terminal includes: a first portion connected to the external electrode and extending in a direction in which the plurality of internal conductors oppose each other; and a second portion extending in a direction intersecting the direction in which the plurality of internal conductors oppose each other.

10. The electronic component according to claim 1, wherein: the metal terminal further includes a metal layer formed on at least one of an upper surface included in the first portion and connecting the first main surface and the second main surface or a lower surface included in the second portion and connecting the first main surface and the second main surface.
